



NAVIGATION

Dashboard

Incident Center

Simulations

Workflows

ACTIVE SERVICES 12 Total

- web-app-v2 98%
- api-server-prod 64%
- postgres-db-01 42%
- redis-cache-lr 100%
- auth-gateway 99%

Settings

SCENARIO: MEMORY LEAK CASCADING FAILURE LATENCY SPIKE

SEED 4392

DIAGNOSIS STEP 04 / 12

RUN PC ALGORITHM

SERVICE HEALTH 3 Active

web-app-v2
SERVICE INSTANCE
CPU USAGE 17.5% MEM RSS 280 MB
P95 LATENCY 165 ms ERROR RATE 0.1/s

api-server-prod ROOT 56%
SERVICE INSTANCE
CPU USAGE 21.8% MEM RSS 384 MB
P95 LATENCY 184 ms ERROR RATE 0.18/s

postgres-db-01 88%
SERVICE INSTANCE
CPU USAGE 27.9% MEM RSS 544 MB
P95 LATENCY 42 ms ERROR RATE 0.05/s

CAUSAL DAG VISUALIZATION

ZOOM 100%

AUTO-LAYOUT

database 88%

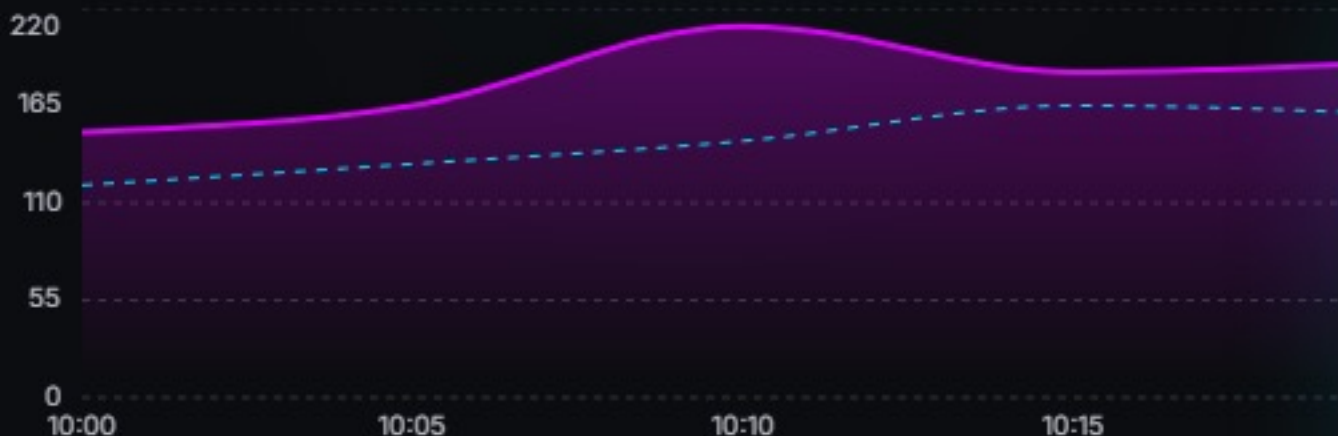
api-server 56%

web-app 55%

METRIC HISTORY

web-app api-server database

Correlated: 0.9



AI INSIGHTS STEP 04

ROOT CAUSE PREDICTION

memory_leak 89%

AI Agent detected abnormal heap growth in **api-server-prod** correlated with increased **database_lowait**.

SHAP FEATURE ATTRIBUTION TOP 5

- api-server memory_rss lag1 11.06
- api-server memory_rss lag3 10.22
- api-server memory_rss lag5 9.35
- api-server memory_rss mean5 8.95
- api-server memory_rss mean15 7.84

Remediation Hub: api-server-prod

Context Active

RESTART NODE

SCALE OUT

DIAGNOSTICS

SIMULATE

DECLARE RESOLUTION

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GLOBAL > SERVICES > API-SERVER-PROD INCIDENT #1942



api-server-prod

WARNING

K8S PODS: 6/6 RUNNING HASH: 7df2a9c UPTIME: 14d 2h 12m

Change Log

Declare Incident

Causal Dependencies

Focused graph: api-server direct paths



LB-NGINX

99%



API-SERVER

64%



POSTGRES

42%

AI Feature Attribution

SHAP breakdown of anomaly score

P99 Latency 85% Impact

Memory RSS 64% Impact

DB Pool Saturation 42% Impact

GC Pauses 28% Impact

CPU Steal 12% Impact

Anomaly Timeline

Latency P99 (ms) vs Predictive Baseline



AI Log Diagnostics

REAL-TIME

[14:21:05] INFO Incoming request to /v1/user/profile from 10.0.4.12
[14:21:08] WARN Postgres connection pool reaching 85% capacity
[14:21:10] ERROR Timeout downstream dependency: auth-service (504 Gateway Timeout)
[14:21:12] INFO Retrying connection to auth-service... attempt 1
[14:21:15] CRIT Fatal leak detected in Heap segment 0x4f2a; 420MB unreferenced

AI-Suggested Remediation

RECOMMENDED

Confidence: 94%

Vertical Scale: api-server-prod

Identified memory leak root cause; scaling will prevent OOM while agent patches code.



Simulation Studio

Execute Fix



ALTERNATIVE

Confidence: 71%

Purge Redis Buffer Cache

Clearing stagnant buffer memory may temporarily alleviate memory pressure.

Test Hypothesis



Manual Diagnostic

Open SSH tunnel to investigate heap dump manually.

Initiate Debug Session



Settings



Counterfactual Simulator

SCENARIO: ALPHA-09

Simulating interventions for **api-server-prod** (US-EAST-1)

INTERVENTION TYPE



Horizontal Scale



Soft Restart



Traffic Drain



Cache Purge

POD COUNT (DESIRED)

8 Instances



ENVIRONMENTAL OVERRIDES

-Xmx8g -XX:+UseG1GC

5000

Compute Counterfactual

PREDICTION OUTCOME

82%

SUCCESS

HIGH PROBABILITY RESOLUTION

CAUSAL EXPLAINER (SHAP)

MEMORY HEADROOM +42%

GC FREQUENCY REDUCTION +28%

NETWORK CONGESTION -12%

CONTEXT SWITCHING -5%

"Increasing pod count directly mitigates the observed memory leak by distributing working sets, reducing garbage collection stop-the-world pauses by an estimated 74%."

TEMPORAL RECOVERY FORECAST (T+30M)

Predicted cost impact: **+\$1.24/hr** infrastructure overhead.Rollback policy: **Active**. Auto-revert if P99 latency spikes.

Discard Scenario

Execute Intervention →

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Remediation Execution

AI AGENT ACTIVE

Executing Vertical Scale-Up for service api-server-prod to resolve P95 Latency Spike (Incident #7742).

View Incident

ABORT & ROLLBACK

Execution Timeline



Snapshot & Backup

14:20:05

Creating volume snapshot for postgres-db-01



Pre-flight Validation

14:20:12

Checking cluster capacity and IAM permissions



Config Patching

14:21:00

Updating HPA minReplicas to 8 for api-server



Traffic Rerouting

Shifting 20% traffic to canary-api-02



Health Verification

Monitoring P95 latency and error rates for 5m



Step 3 of 5: Config Patching

Patching Deployment Specs...

45%

OVERALL PROGRESS

REMEDIATION SCRIPT OUTPUT

```
01 [14:20:01] [INFO] Initializing Remediation Workflow: scale_api_replicas
02 [14:20:05] [INFO] Task: SNAPSHOT_STORAGE status: SUCCESS
03 [14:20:12] [INFO] Task: VALIDATE_K8S_CONTEXT status: SUCCESS
04 [14:21:00] [ACTION] Patching deployment/api-server in namespace production
05 [14:21:05] [INFO] kubectl patch hpa api-server --patch '{"spec":{"minReplicas":8}}'
06 [14:21:10] [WAIT] Waiting for 3 additional pods to reach READY state...
07 [14:21:15] [INFO] Pod api-server-7f8d9b-x2k1 status: RUNNING
08 [14:21:18] [INFO] Pod api-server-7f8d9b-l9p4 status: RUNNING
09 [14:21:20] [INFO] Pod api-server-7f8d9b-m5q1 status: CONTAINER_CREATING
10
```

Impact Verification

LIVE LATENCY TARGET BASELINE



P95 Latency

IMPROVED

180ms

520ms ↘ -65%

Error Rate

IMPROVED

0.15%

2.4% ↘ -94%

Remediation Metadata

Action Type

VerticalScale

Causal Confidence

98.4%

Cluster Zone

us-east-1a

AI STRATEGY NOTE

"Scaling memory limit from 2Gi to 4Gi based on SHAP value attribution pointing to memory thrashing in api-server-prod. Estimated recovery to baseline: 184s."